



Bore measuring tool

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Rebuild (CRC)**

**MARC
Management**

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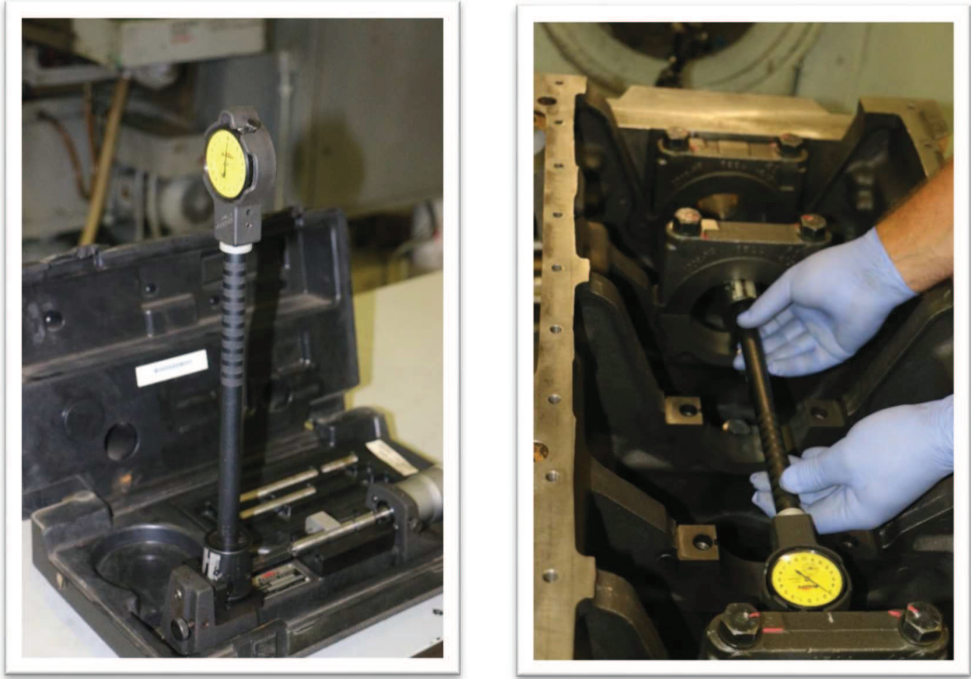
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September 2016
4.01-160907-01

1.0 Introduction

During engine cylinder block repair, there is a necessity to machine the main bore. In order to measure the diameter, it is recommended to use the following Caterpillar tools: 6V7899 and 6V7898. During boring it is necessary to remove the boring bar and measure the bore. That leads to time loss.

Usual practice:



Amur Machinery has made a tool that allows a measurement of the bore diameter without removing the tooling. This reduces the time and makes machining the bore more accurate. Using this tooling you can take measurements frequently during the boring operation.



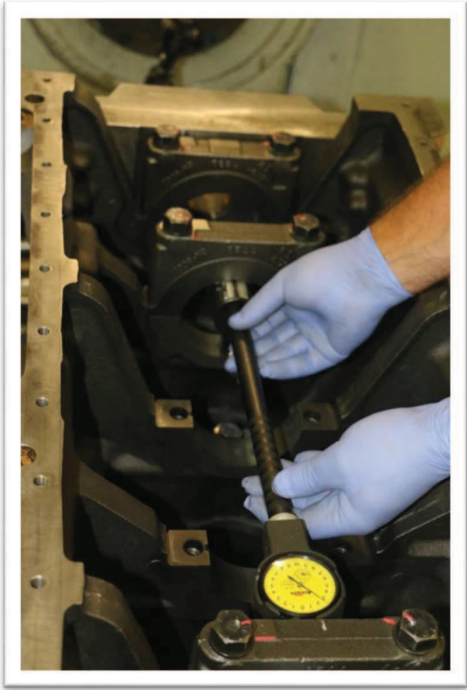
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2.0 Best Practice Description

The equipment consists of a dial test indicator (DTI) and a U-type body that allows you to measure the diameter when the boring bar is inside the bore. Below are before-after photos of a measure process.

Before-After photos of measure process

Before



After



3.0 Implementation Steps

- Design the tool for accurate and simple measure.
- To make such a device we use details of the bore measurement tools:

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1. Measuring tool



2. Dial test indicator (DTI)

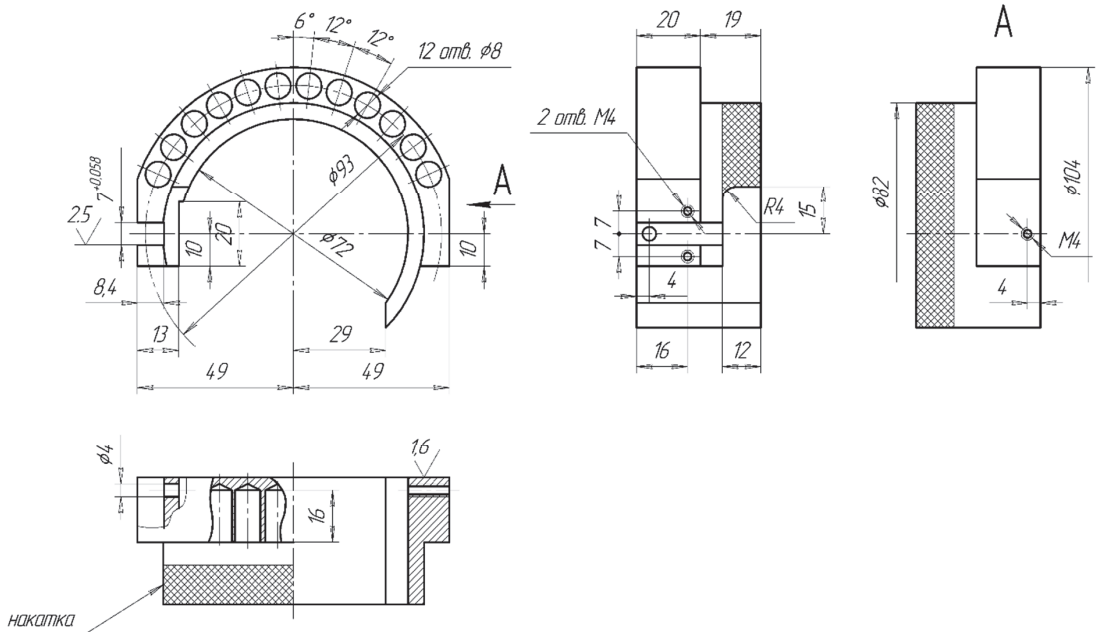


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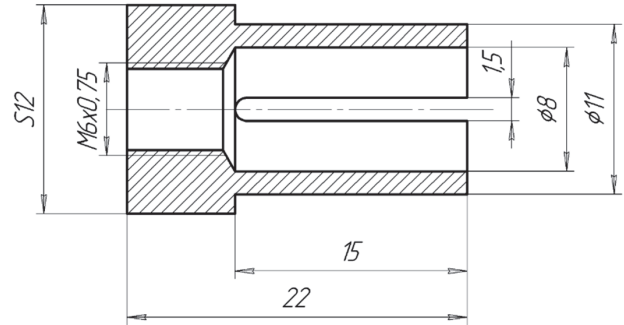
3. Then there were made the following details:

- U - type body

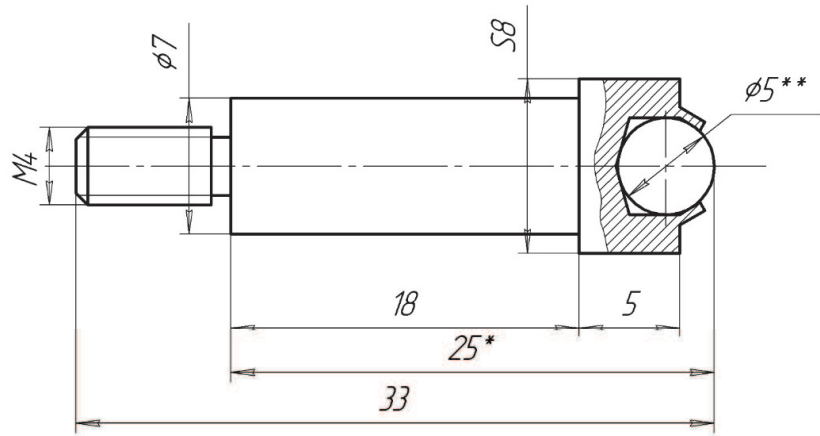


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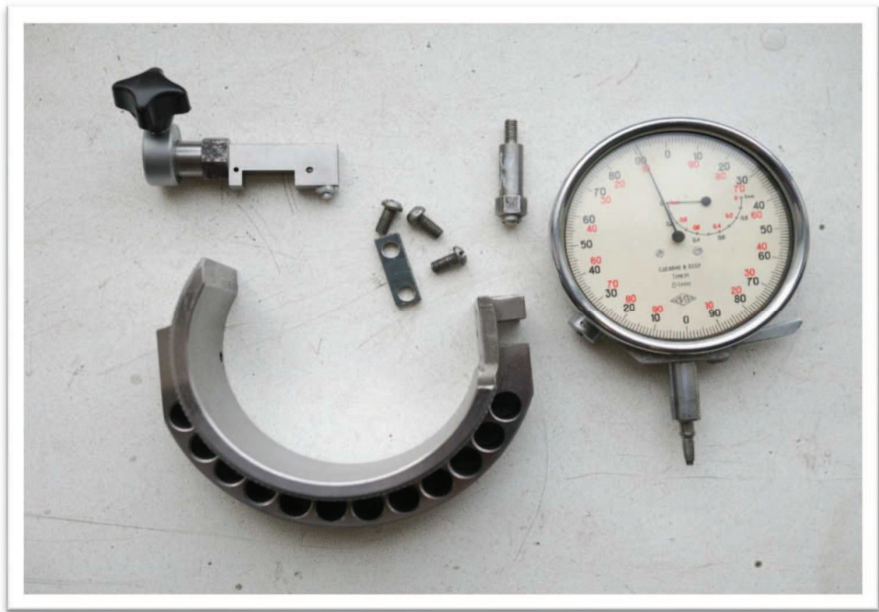


• Pin



* for measuring bores with 130 mm diameter
** steel ball

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After assembly the tool is ready to use.



The tool is used for bore measurement with diameters from 115 to 130 mm

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4.0 Benefits

- Accurate measure and boring;
- Reduces time for disassembly and installation of the tool up to 20%;
- The size and the base for a measure tool makes it easier to store it.

5.0 Resources Required

Minimal expense to produce the tool.

6.0 Supporting Attachments / References

N/a

7.0 Related Best Practices

N/a.

8.0 Acknowledgements

The idea of creating & implementing this Best Practice belongs to:

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